

Experiment 19:

Implementation of Naive Bayes classification for Bank Loan prediction

Python Code:

```
import numpy as np
from sklearn.naive_bayes import CategoricalNB
X = np.array([[1,1,0],[1,0,0],[0,1,1],
              [0,0,1],[1,1,1],[0,0,0]])
y = np.array([1,1,0,0,1,0])
model = CategoricalNB()
model.fit(X, y)
sample = [[1,1,0]]
prediction = model.predict(sample)[0]
print("Dataset:")
print("Income Credit_History Loan_Status")
for i in range(len(X)):
    print(f'{X[i,0]:6} {X[i,1]:14} {'Approved' if y[i] else 'Rejected'}')
print("\nLoan Prediction:", "Approved" if prediction else "Rejected")
```

Output:

```
Dataset:
Income  Credit_History  Loan_Status
    1           1      Approved
    1           0      Approved
    0           1      Rejected
    0           0      Rejected
    1           1      Approved
    0           0      Rejected
```

```
Loan Prediction: Approved
```